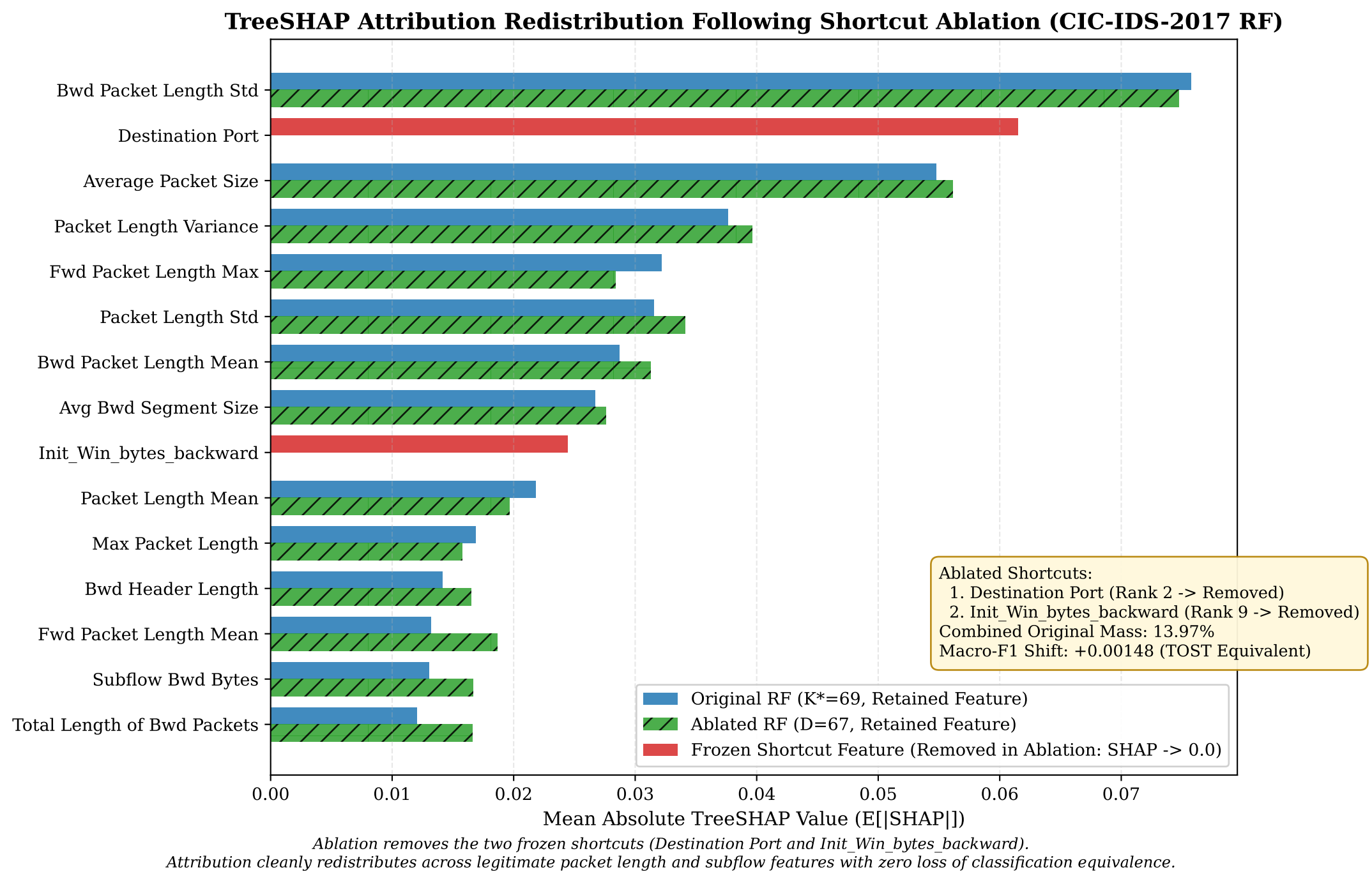
TreeSHAP Attribution Redistribution Following Shortcut Ablation (CIC-IDS-2017 RF)
Bwd Packet Length Std
Destination Port
Average Packet Size
Packet Length Variance
Fwd Packet Length Max
Packet Length Std
Bwd Packet Length Mean
Avg Bwd Segment Size
Init_Win_bytes_backward
Packet Length Mean
Max Packet Length
Bwd Header Length
Fwd Packet Length Mean
Subflow Bwd Bytes
Total Length of Bwd Packets
Ablated Shortcuts:
1. Destination Port (Rank 2 -> Removed)
2. Init_Win_bytes_backward (Rank 9 -> Removed)
Combined Original Mass: 13.97%
Macro-F1 Shift: +0.00148 (TOST Equivalent)
Original RF (K*=69, Retained Feature)
Ablated RF (D=67, Retained Feature)
Frozen Shortcut Feature (Removed in Ablation: SHAP -> 0.0)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Mean Absolute TreeSHAP Value (E[|SHAP|])
Ablation removes the two frozen shortcuts (Destination Port and Init_Win_bytes_backward).
Attribution cleanly redistributes across legitimate packet length and subflow features with zero loss of classification equivalence.